Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

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Advanced Algorithms and Parallel Programming – II part – September 5th 2024

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

Answer the following questions about parallel application modeling and briefly explain (without an explanation, the answer will be considered invalid)

A. Please comment on this sentence: "Automatic parallelization cannot be used because it produces wrong solutions.". True/false, partially true? (1)

B. "A single instruction can manipulate different data at a time". Please comment on the sentence and discuss which kind of parallelism it is referring to. (1)

C. Which loop can be parallelized and under which conditions? Does the overhead of the runtime impact the parallelization of the loops? Which is the best way to parallelize this code on multi-core systems? (2)

```
for (i=0; i<1000; i++)
        for (j=0; j<4; j++)
            a[i][j] = f(a[i][j]) + a[i][j];
```

Exercise n. 2

Answer the following questions about parallel patterns and briefly explain (without an explanation, the answer will be considered invalid)

A. Please describe the parallel reduce pattern: assumptions, complexity, and example. (2)

B. Please explain how the parallel shift pattern functions and provide its implementation (including pseudo-code). Also, please indicate its complexity. (2)

Exercise n. 3

Answer the questions about the shared memory parallel programming and briefly explain (without an explanation, the answer will be considered invalid)

A. Describe the method to qualitatively evaluate a parallel algorithm through its data flow graph. (1)

B. Describe what is the effect of the following OpenMP directive: (2)

```
#pragma omp for simd schedule(simd:static, 5)
```

C. In Halide, it is good practice to schedule pipeline stages from the output to the input. Is this statement true or false? Explain. (1)

Exercise n. 4

Answer the following questions about MPI and briefly explain (without an explanation, the answer will be considered invalid)

A. Provide an example of when it is useful to use the MPI_Barrier function. (1) **This question can be skipped if you passed the challenge.**

B. What is the main difference between MPI and OpenMPI? (1)

C. Since each MPI process has a different memory address space, how is it possible to have a deadlock? Also, make an example to explain how we can solve it. (2)